

Section A

For each question there are four possible answers, **A**, **B**, **C** and **D**. Choose the **one** you consider to be correct.

- 1 In the discovery of an element, **E**, scientists found that it consist of 3 isotopes, ^{140}E , ^{142}E and ^{144}E in the atomic ratio of 2: 3: 1. Calculate the relative atomic mass of element **E** to 1 decimal place.

- A** 140.5
- B** 141.7
- C** 142.0
- D** 143.1

$$\text{Ar of E} = \frac{(140 \times 2) + (142 \times 3) + (144 \times 1)}{(2+3+1)} = \frac{850}{6} = 141.67 = \underline{\underline{141.7}} \text{ (1 d.p)}$$

Ans: B

- 2 To determine the formula of an unknown hydrocarbon, **Y**, a student exploded 20 cm³ of **Y** in 150 cm³ of oxygen. At the end of the reaction, an expansion of 20 cm³ was recorded, which contracted by 80 cm³ when cooled. The resultant residual gas was then passed through an alkali which resulted in a further contraction to 10 cm³. What is the formula of the unknown hydrocarbon **Y**?

- A** C₄H₈
- B** C₄H₁₀
- C** C₅H₈
- D** C₅H₁₀

Equation for the general combustion of hydrocarbon:



Since there is a contraction of 80 cm³ when cooled after combustion,

80 cm³ = volume of water produced

Thus, ratio of hydrocarbon : water = 20 : 80 = 1 : 4 and (y/2) = 4 and **y = 8**

Total volume of gas before combustion = 170 cm³

Total volume of gas after combustion = 190 cm³ (expansion of 20 cm³)

Thus after contraction of water (80 cm³) = 190 – 80 = 110 cm³

After passing through alkali, volume of gas reduced to 10 cm³

Thus, volume of CO₂ produced = 110 – 10 = 100 cm³

Ratio of hydrocarbon : CO₂ = 20 : 100 = 1: 5 and **x = 5**

Formula of hydrocarbon = C₅H₈ and **Ans: C**

- 3 In a redox reaction, 0.635 g of pure copper, Cu, required 20.00 cm³ of 1.00 mol dm⁻³ nitric acid, HNO₃, for complete reaction. If Cu²⁺ ions was produced at the end of the reaction, what is the oxidation number of nitrogen in the products formed?
- A +1
B +2
C +3
D +4

No. of moles of Cu = $0.635 / 63.5 = 0.01$

No. of moles of HNO₃ = $(20/1000) \times 1 = 0.02$; Oxidation number of N = +5

Thus, Cu : HNO₃ = 0.01 : 0.02 = 1 : 2

Since Cu²⁺ is produced from Cu, 2 electrons are produced per mole of Cu

These 2 electrons are taken in by 2 moles of HNO₃ (based on mole ratio calculated above)

Thus 1 mole of HNO₃ taken in 1 mole of electron and its also reduced (since Cu is oxidised)

Therefore the new oxidation number of HNO₃ = $5 - 1 = +4$

Ans: D

- 4 An element **G** in Period 2 has a general valence electronic configuration of ns² np³. What is the formula of the fluoride that is most likely to be formed by element **G**?
- A GF₂
B GF₃
C GF₄
D GF₅

With a general electronic configuration of ns² and np³, it belongs to Group V and its nitrogen.

Thus fluoride formed by N is NF₃ since it can only accommodate 8 outermost shell electrons as it belongs to period 2.

Ans: B

5 Use of the Data Booklet is relevant to this question.

The table below shows statements that 2 students made about the Ca^{2+} and Fe^{2+} ions.

student	statement
Q	Both ions were obtained by removing the electrons from the same orbital.
R	Both ions have more protons than neutrons and more protons than electrons.

Which students are correct?

- A Student **Q** only
- B Student **R** only
- C Both Student **Q** and **R**
- D None of them

Ca: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$

Ca^{2+} : $1s^2 2s^2 2p^6 3s^2 3p^6$ (electrons removed from 4s orbital)

(protons = 20, electrons = $20 - 2 = 18$, neutrons = $40.1 - 20 = 20.1$)

Fe: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^6 4s^2$

Fe^{2+} : $1s^2 2s^2 2p^6 3s^2 3p^6 3d^6$ (electrons removed from 4s orbital)

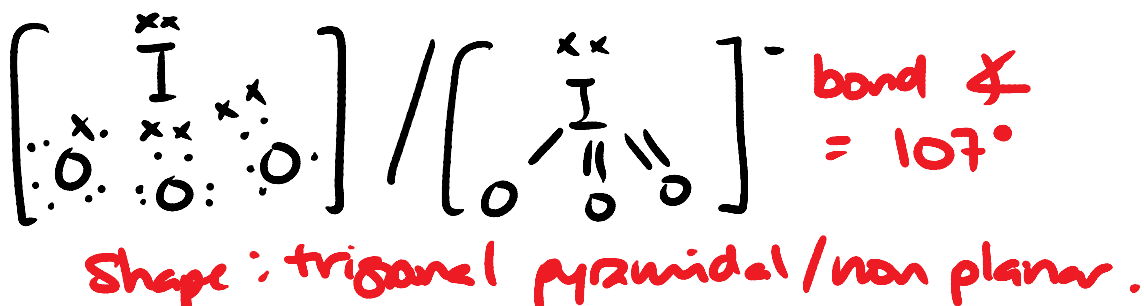
protons = 26, electrons = $26 - 2 = 24$, neutrons = $55.8 - 26 = 29.8$)

Thus only student Q is correct; student R is wrong as both have more neutrons than protons.

Ans: A

6 Iodic acid is a strong acid that exist as a white solid at room temperature. It has a chemical formula of HIO_3 and dissolves readily in water to give H^+ and IO_3^- ions. Which of the following statements about the ion IO_3^- is true?

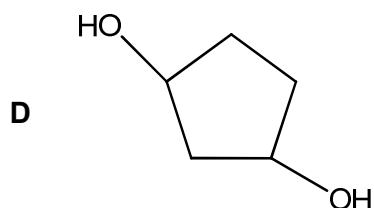
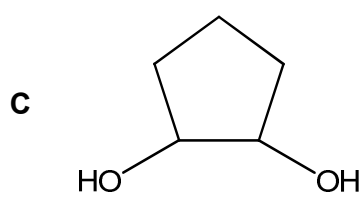
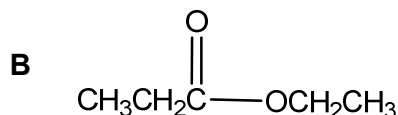
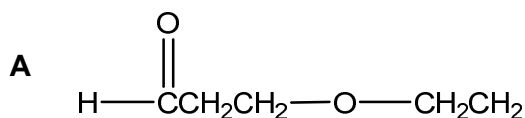
- A It is a non-planar ion.
- B It has a bond angle of 120° .
- C Only sigma bonds are formed between iodine and oxygen in the ion.
- D The oxidation number of iodine in the ion is negative.



- Bond between iodine and oxygen are both sigma and pi (due to double bond formed)
- oxidation number of iodine (x) in ion: $-6 + x = -1$ Thus $x = +5$

Ans: A

- 7 The following compounds are isomers with molecular formula $C_5H_{10}O_2$. Which of them is the least volatile?



Intermolecular forces for

A: VDWs

B: VDWs

C: Hydrogen bonding

D: Hydrogen bonding

Thus, C and D would have a higher boiling point (bp) than A and B

However, between C and D, C has intramolecular hydrogen bonding due to proximity of the 2 OH groups resulting in a less extensive intermolecular hydrogen bonding in C.

Thus, D would have the highest bp among the 4 options making it the least volatile.

Ans: D

- 8 Graphite is a common form of carbon that has many uses. One of them is as lubricants

[Turn over

in machine parts. Which of the following properties of graphite allow for it to be used as lubricants?

- A** Each carbon atom in graphite is sp^2 hybridised.
- B** Graphite has a very high melting point.
- C** The interactions between layers of graphite is van der Waals forces of attraction.
- D** Graphite has an extended π -electron cloud above and below each plane of carbon atoms.

The property that allow for graphite to be a lubricant is due to the fact that it is **soft** and the layers can **slide pass each other** and this is due to the fact that the interactions between the layers in graphite is weak VDWs forces of attraction.

Ans: C

- 9** Which of the following arranges the ionic salts in increasing magnitude of their lattice energy?

- A** $Al_2O_3 < MgO < MgBr_2 < NaBr$
- B** $NaBr < MgO < MgBr_2 < Al_2O_3$
- C** $MgO < NaBr < Al_2O_3 < MgBr_2$
- D** $NaBr < MgBr_2 < MgO < Al_2O_3$

The magnitude of lattice energy is dependant/proportionate to $(q^+)(q^-)/(r^+)+(r^-)$

From the formula, one can see that

- it is dependant on both charge of the ions and the ionic radius.
- charge would take higher priority over ionic radius since in the formula, charge is multiplied but ionic radius is just the sum or addition.
- The magnitude also varies proportionately to charge but inversely proportionate to ionic radius

Thus, Al_2O_3 will have the largest magnitude since the product of the charges is the largest ($3 \times 2 = 6$)

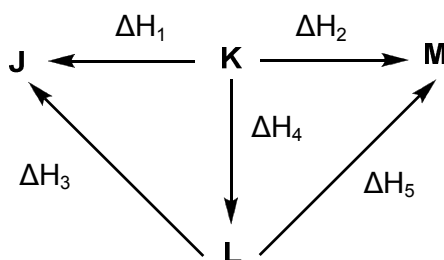
This is followed by MgO , whose product of the charges is the second largest ($2 \times 2 = 4$)

This is then followed by $MgBr_2$ ($2 \times 1 = 2$)

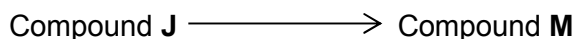
The smallest magnitude is by $NaBr$ whose product of the charges is ($1 \times 1 = 1$)

Ans: D

- 10 The energy cycle below shows the reaction pathways between Compounds **J – M**.



What is the enthalpy change for the following reaction?



- A** $\Delta H_1 + \Delta H_2$
B $\Delta H_2 - \Delta H_3 - \Delta H_4$
C $\Delta H_3 + \Delta H_5$
D $\Delta H_4 - \Delta H_1 - \Delta H_5$

By Hess Law, going starting with compound J and ending with compound M,
 We can go from compound J to L ($-\Delta H_3$ as going against the arrow), then from L to K ($-\Delta H_4$ as going against the arrow) and finally from K to M ($+\Delta H_2$ as going with the arrow)
 Thus, ΔH for cpd J to M = $-\Delta H_3 - \Delta H_4 + \Delta H_2 = \Delta H_2 - \Delta H_3 - \Delta H_4$

Ans: B

- 11 To study the kinetics of a reaction between **X**, **Y** and **Z**, a student conducted a series of experiments and the following results were obtained:

Experiment	Vol of X / cm ³	Vol of Y / cm ³	Vol of Z / cm ³	Vol of H ₂ O / cm ³	Relative rate
1	20	15	10	55	1
2	40	15	10	35	4
3	40	25	15	20	6
4	40	15	20	25	8

What is the rate equation for the reaction?

- A** Rate = $[X]^2[Z]$
B Rate = $[X]^2[Y]$
C Rate = $[X][Y][Z]$
D Rate = $[X]^2[Y][Z]$

Since its volume and the total volume used for all experiment is 100, volume can be used to compare directly as volume is proportionate to concentration.

Comparing expt 1 & 2,

When $[X]$ doubled, rate increase by 4 times. Hence order of reaction w.r.t. X is 2

Comparing expt 2 & 4,

When $[Z]$ doubled, rate increase by 2 times. Hence order of reaction w.r.t. Z is 1

Comparing expt 2 & 3 and based on what was calculated previously,

$$\text{rate}_3/\text{rate}_4 = [k(40)^2(15)^y(10)] / [k(40)^2(25)^y(15)]$$

$$4/6 = (10/15)(15/25)^y$$

$$2/3 = (2/3)(3/5)^y$$

$$(3/5)^y = 1$$

$$y = 0$$

Thus rate equation: $\text{rate} = k[X]^2[Z]$

Ans: A

- 12 A technician found the following information in a handbook about an equilibrium reaction that the factory carries out on an industrial scale.

Temperature / °C	Kc
250	1×10^{-5}
500	1×10^{-4}

Pressure / atm	% Product
5	33
10	47

Based on the above information, what can be said about the equilibrium reaction that the factory carries out?

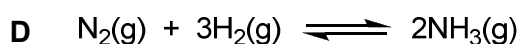
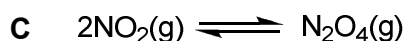
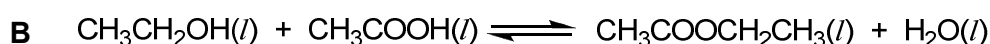
- A The forward reaction is endothermic and there are more gaseous molecules in the products than reactants.
- B The forward reaction is exothermic and there are less gaseous molecules in the products than reactants.
- C The forward reaction is endothermic and there are less gaseous molecules in the products than reactants.
- D The forward reaction is exothermic and there are more gaseous molecules in the products than reactants.

When temperature increased, K_c increased. This means the formation of the product is favoured when temperature increase. That would mean that the forward reaction must be endothermic (to absorb the excess heat)

When pressure increase, % of product also increased. This means the product side would have less no. of gaseous molecules in order to counter the increase in pressure.

Ans: C

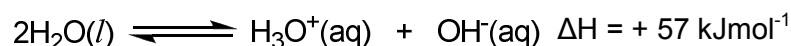
- 13 Which of the following equilibrium equations have no units for their equilibrium constant, K_c ?



Only option B includes all the species reflected in the equation in the K_c and the number of moles of reactants is the same as the products resulting in the units in the product cancelling the units of the reactants and hence no units for K_c .

Ans: B

- 14 The equation for the dissociation of water is as follows:



Which of the following statements about the above equilibrium is correct?

- A As temperature increases, the pH of water also increases.
- B As the volume of water increases, the pH of water also increases.
- C The summation of pH and pOH is 14 at all temperatures.
- D $[\text{H}^+]$ will always equal to $[\text{OH}^-]$ at all temperatures.

A: When temperature increase, the forward reaction would be favoured. This would result in $[\text{H}^+]$ to also increase. This would result in pH to decrease and not increase.

B: $[\text{H}_+]$ would remain constant even when volume of water increase (as both no. of moles of water and volume increase proportionately). Hence pH would be constant and not increase.

C: pH + pOH is only 14 at 25°C

D: This one is true as the dissociation of water at any temperature would always yield the same no. of moels of H^+ and OH^- .

Ans: D

- 15** Given that both NaOH and KOH are strong mineral bases, calculate the resultant pH when 20 cm³ of 0.300 moldm⁻³ of NaOH is mixed with 30 cm³ of 0.200 moldm⁻³ of KOH.

- A** 11.8
B 12.1
C 13.4
D 13.7

No. of moles of OH⁻ from NaOH = (20/1000) × 0.3 = 0.006

No. of moles of OH⁻ from KOH = (30/1000) × 0.2 = 0.006

Total concentration of OH⁻ = (0.006 × 2) / [(20+30)/1000] = [0.012/(50/1000)]

= 0.24 moldm⁻³

Thus, pOH = -log(0.24) = 0.620 and pH = 14 – 0.620 = 13.4

Ans: C

- 16** Which of the following properties show a decreasing trend across the elements of Period 3?

- A** pH of their chlorides in water
B Melting point of their oxides
C Electrical conductivity
D Ionic radius

A: It decreases across all elements in Period 3.

B: Melting point actually increases first from Na₂O to MgO before decreasing.

C: This one also increases from Na to Al before decreasing.

D: There is an increase from Si to P

Ans: A

- 17** An element in Period 3 has the following properties:

- It does not conduct electricity.
- Its first ionisation energy is larger than both the elements before and after it.

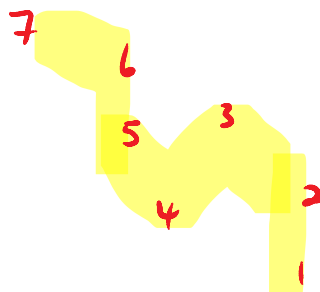
Which of the following statements about this element is correct?

- A** It has a high melting and boiling point.
B It forms chlorides that are mildly acidic.
C It forms oxides that are very acidic.
D Its oxidation state in compounds usually follows the group number it belongs to in the Periodic Table.

Based on the information given in the question, the element has to be phosphorus.
Hence, based on the property of P, the only one that is true is option C

Ans: C

- 18 What is the IUPAC name of the skeletal structure below?



- A 2-ethyl-5,5-dimethylpentane
- B 2-ethyl-5-methylhexane
- C 1,2,5-trimethylhexane
- D 2,5-dimethylheptane

The longest carbon chain is as highlighted in the question.

Since longest carbon chain is 7 carbon, the backbone of the molecule is heptane

The methyl group is on carbon 2 and 5

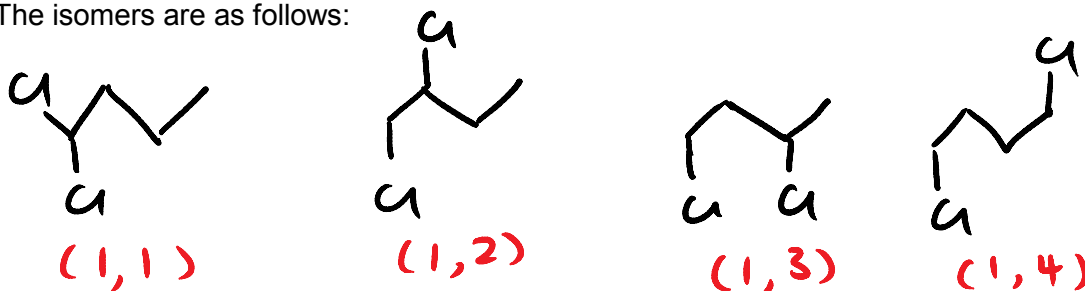
Hence IUPAC name is 2,5-dimethylheptane

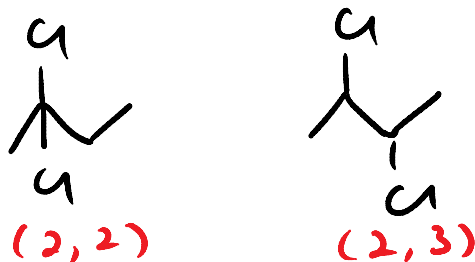
Ans: D

- 19 How many positional isomers are there for a dichlorobutane?

- A 5
- B 6
- C 7
- D 8

The isomers are as follows:





Ans: B

- 20 Ethene is shaken with an aqueous mixture of iodine and sodium bromide.

Which of the following statements about the above reaction is **incorrect**?

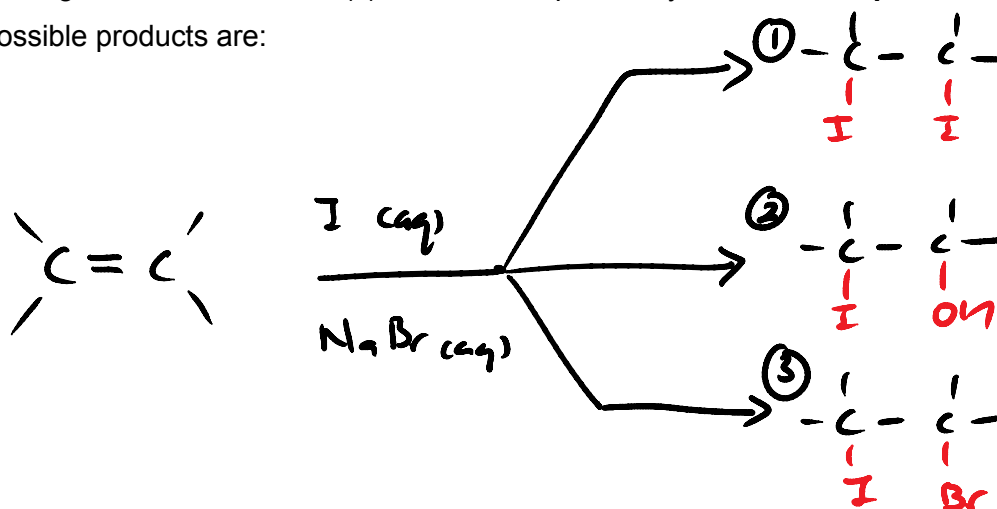
- A There will at least one bromine atom found in all of the products formed.
- B There will be a mixture of three different products formed.
- C This is an addition reaction.
- D The role of iodine in this reaction is an electrophile.

This is an **addition reaction** between ethane and iodine.

Hence, **iodine is the electrophile** that will be attacked by the double bond

The OH^- and Br^- from NaBr will only be added to the intermediate to complete the reaction together with iodide ion (I^-) and hence a possibility of **3 different products**

The possible products are:



Ans: A

- 21 Which of the following properties of benzene is **not** associated with the ring of delocalised electrons?

- A Benzene tend to attract electrophiles rather than nucleophiles.
- B Benzene tend to undergo substitution reactions rather than addition reactions
- C Conversion of benzene to cyclohexane requires very harsh conditions.
- D Benzene is a planar molecule with all the carbon atoms lying on the same plane.

A: Due to the ring of delocalised electrons, they would attract electrophiles, which are electron deficient species

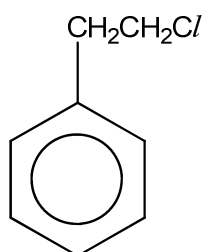
B: A benzene reaction would always want to re-establish the ring of delocalised electrons and hence undergo substitution rather than addition that would destroy the ring of delocalised electrons

C: The delocalised electrons help to stabilize benzene and hence any reaction that destroys the ring of delocalised electrons will need very harsh conditions

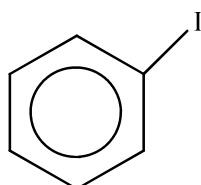
D: Although true, benzene is planar has nothing to do with the ring of delocalised electrons but due to the no. of electrons and lone pair each C atom have

Ans: D

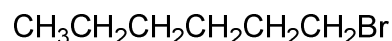
22 The following are three different types of halogen derivatives:



1-chloro-2-phenylethane



iodobenzene



1-bromohexane

A student then added ethanolic AgNO_3 separately to the 3 halogen derivatives above.

Which of the following shows the correct order in which the precipitate appears, from the fastest to the slowest?

- A** 1-bromohexane, iodobenzene, 1-chloro-2-phenylethane
- B** 1-bromohexane, 1-chloro-2-phenylethane, iodobenzene
- C** iodobenzene, 1-bromohexane, 1-chloro-2-phenylethane
- D** 1-chloro-2-phenylethane, 1-bromohexane, iodobenzene

The reaction mentioned in the questions for RX molecules involve the breaking of the C-X (halogen) bond. The strongest is the iodine attached to benzene and hence it would be the slowest to show the ppt.

Between the first and the 3rd molecule, the C-Br bond is weaker and hence would break faster and hence show the ppt the fastest.

Hence the order from the fastest to slowest should be molecule 3, 1 and then 2.

Ans: B

23 Which of the following reagents and conditions, when added to ethanol, involve the cleavage of the O-H bond in ethanol?

- A** CH_3COOH , excess concentrated H_2SO_4 and reflux
- B** I_2 dissolved in NaOH and warm
- C** PCl_5 at room temperature
- D** $\text{K}_2\text{Cr}_2\text{O}_7$, $\text{H}_2\text{SO}_4(\text{aq})$ and heat

A: It involved the breaking of the O-H bond in the alcohol

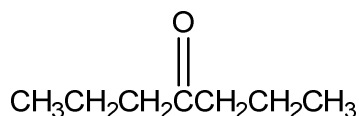
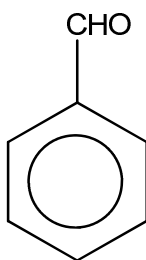
B: It breaks the C- CH_3 bond in ethanol

C: It breaks the C-O bond in ethanol.

D: It involves the breaking of the bonds of the C with the OH group but does not break the O-H bond.

Ans: A

24 Which of the following reagents can be used to distinguish between the following 2 compounds?



- A** 2,4-dinitrophenylhydrazine
- B** Fehling's solution
- C** Tollen's reagent
- D** LiAlH_4 in dry ether

A: Will react with both as both are carbonyl compounds

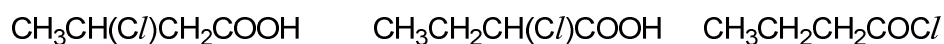
B: Will not react with both of them as both are non-aliphatic aldehydes

C: Will only react with benzaldehyde but not the ketone

D: Will react with both and reduce both to alcohols.

Ans: C

- 25** Below are 3 compounds of carboxylic acids and its derivatives.



Which of the following arranges the pH of their solutions in increasing order when they are placed in water?

- A** $\text{CH}_3\text{CH}(\text{Cl})\text{CH}_2\text{COOH}$, $\text{CH}_3\text{CH}_2\text{CH}(\text{Cl})\text{COOH}$, $\text{CH}_3\text{CH}_2\text{CH}_2\text{COCl}$
B $\text{CH}_3\text{CH}_2\text{CH}_2\text{COCl}$, $\text{CH}_3\text{CH}(\text{Cl})\text{CH}_2\text{COOH}$, $\text{CH}_3\text{CH}_2\text{CH}(\text{Cl})\text{COOH}$
C $\text{CH}_3\text{CH}_2\text{CH}_2\text{COCl}$, $\text{CH}_3\text{CH}_2\text{CH}(\text{Cl})\text{COOH}$, $\text{CH}_3\text{CH}(\text{Cl})\text{CH}_2\text{COOH}$
D $\text{CH}_3\text{CH}(\text{Cl})\text{CH}_2\text{COOH}$, $\text{CH}_3\text{CH}_2\text{CH}_2\text{COCl}$, $\text{CH}_3\text{CH}_2\text{CH}(\text{Cl})\text{COOH}$

The third one is an acid chloride and hence when dissolves in water to give HCl, a very strong mineral acid and hence the most acidic with the lowest pH.

Between the first 2, the 2nd molecule has the Cl group nearer the COOH group. Being a electron-withdrawing group, the Cl nearer the COOH group help to better disperse the negative charge in the conjugate base in the 2nd molecule than the 1st

Hence the 2nd molecule is more acidic with a lower pH than the 1st molecule.

Hence in increasing order of pH, it should be the 3rd molecule, followed by the 2nd and then the 1st.

Ans: C

Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

- 26** BeCl_2 is able to react with PCl_3 to form a stable product.

Which of the statements about this reaction are correct?

- 1** The mole ratio of BeCl_2 : PCl_3 for this reaction is 1 : 2.
- 2** The product in this reaction is able to form because beryllium has energetically accessible orbitals to expand its octet configuration.
- 3** The bond angle about phosphorus does not change during this reaction.

1: True as BeCl_2 has only 4 outer most shell electrons around Be and hence have 2 empty orbitals to accept 4 more electrons to fulfil its octet rule. Since PCl_3 only has one lone pair, 2 moles of PCl_3 is needed and hence the ratio of the reaction between BeCl_2 and PCl_3 is 1: 2.

2: Not true as Be is in period 2 and hence unable to expand its octet.

3. Not true as a new bond is formed between P and Be in the reaction, the bond angle around P is bound to change.

Ans: D

- 27** A student investigated a reaction between compounds **T**, **U** and **V** and found that the order of reaction is first order with respect to **T**, zero order with respect to **U** and second order with respect to **V**. When a graph was drawn, a straight line was obtained.

Which of the following could be the axis of this graph?

- 1** Rate against **[T]**
- 2** Rate against **[U]**
- 3** Rate against **[V]²**

1: True as the rate is first order w.r.t **T**, you will get a straight line through the origin

when this graph is plotted.

2: True as the rate is zero order w.r.t **U**, you will get a straight vertical line when this graph is plotted

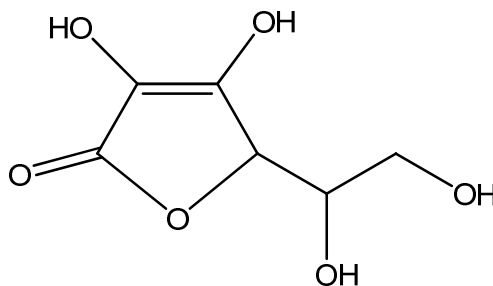
3: True as the rate is second order w.r.t **V**, you will get a straight line through the origin when this graph is plotted.

Ans: A

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

- 28** Vitamin C is a common compound found in many citrus fruits such as oranges and pineapples. It has the following structure:



Which of the following statements about Vitamin C is *incorrect*?

- 1** It is able to exhibit geometric isomerism.
- 2** It will give orange crystals with Brady's reagent.
- 3** 1 mole of Vitamin C will react with excess sodium to give 2 moles of hydrogen gas.

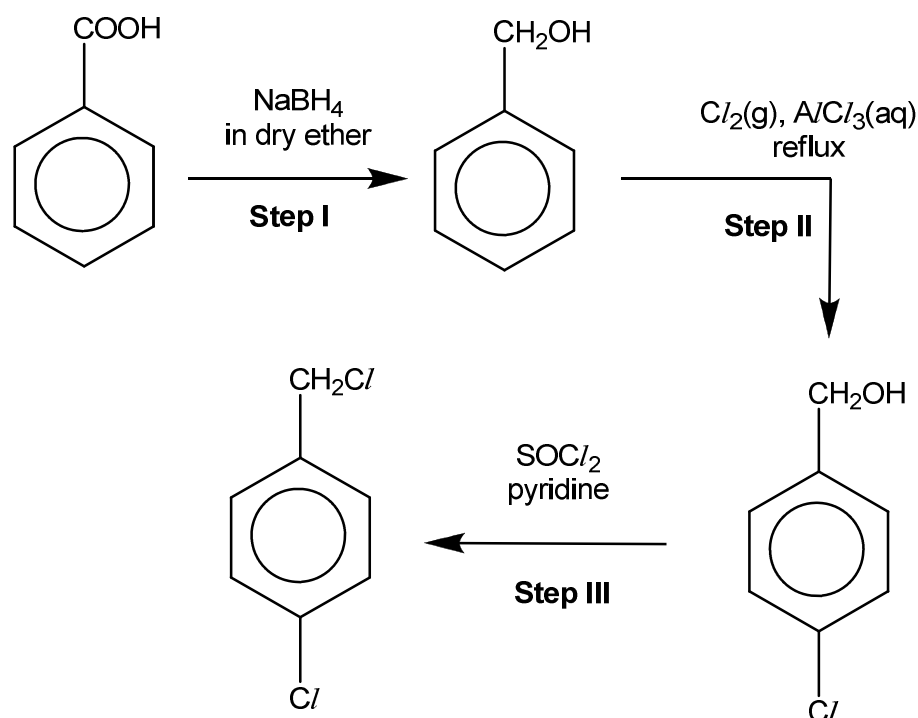
1: Its incorrect (and hence an answer) as the molecule does not exhibit geometric isomerism as it does not have an aliphatic alkene or cyclic alkane with different groups.
 2: Its incorrect (and hence an answer) as it does not contain any carbonyl group.
 3: Its correct (and hence not an answer) as it will react with all 4 OH groups and hence give 2 moles of H₂ gas.

Ans: B

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

29 The following synthetic pathway was proposed by a student:



Which of the following steps are unable to give the desired product that the student proposed in this synthesis?

- 1 Step I**
- 2 Step II**
- 3 Step III**

Step 1: Unable to as NaBH_4 is unable to reduce the acid group, only LiAlH_4 is able to.

Step 2: Unable to work as well as the AlCl_3 is (aq) which will destroy it and hence not perform its role as a catalysis to generate the Cl^+ electrophile.

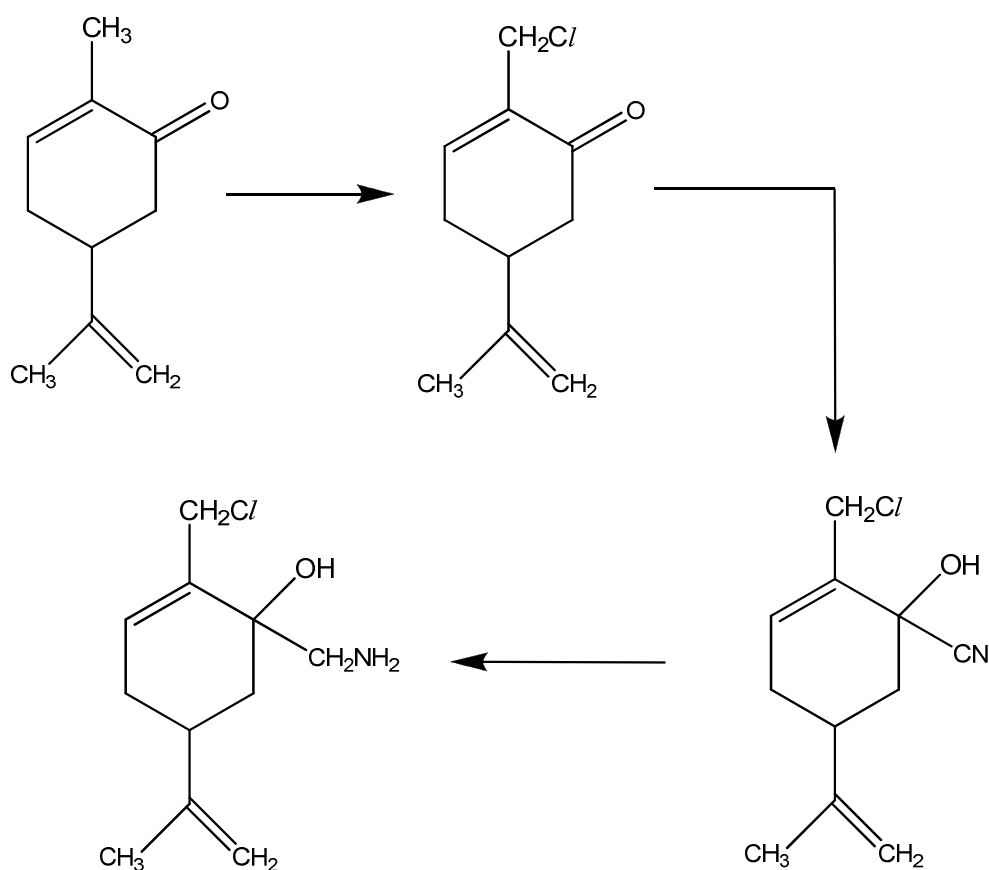
Step 3: Able to work as it's a reagent to substitute the OH group with a Cl group.

Ans: B

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

- 30** Below is the synthesis pathway of Carvone, the main flavouring compound in spearmint oil.



Which of the following types of reactions are shown in the synthetic pathway above?

- 1** Addition
- 2** Reduction
- 3** Substitution

1: Step 2 is an addition reaction of the ketone group.

2: Step 3 is a reduction reaction of the CN group.

3: Step 1 is a substitution reaction being a free radical substitution.

Ans: A

- THE END -

[Turn over

Answers:

1	B	6	A	11	A	16	A	21	D	26	D
2	C	7	D	12	C	17	C	22	B	27	A
3	D	8	C	13	B	18	D	23	A	28	B
4	B	9	D	14	D	19	B	24	C	29	B
5	A	10	B	15	C	20	A	25	C	30	A